

Semi-inducibility of Alternating Cycles of Length $4k + 2$

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Abstract

Let $W: [0, 1]^2 \rightarrow [0, 1]$ be a graphon, and let $\text{Alt}_{2m}(W)$ denote the density of the red–blue alternating cycle of length $2m$, where red edges contribute W and blue edges contribute $1 - W$. We prove that, for every odd integer $m \geq 3$,

$$4^m \text{Alt}_{2m}(W) + t(C_{2m}, 2W - 1) \leq 1.$$

Since the signed even-cycle density $t(C_{2m}, 2W - 1)$ is nonnegative, this implies

$$\text{Alt}_{2m}(W) \leq 4^{-m}.$$

Equivalently, for every $k \geq 1$, the alternating cycle of length $4k + 2$ has semi-induced density at most $2^{-(4k+2)}$. Equality holds only for the constant graphon $W \equiv 1/2$. The proof reduces the problem to an abstract rank-one matrix inequality. A rank-two Schur-complement factorisation produces a scalar power series whose coefficients form a nonincreasing nonnegative sequence; since m is odd, comparing coefficients in the log-determinant expansion gives the required sign.

1 Definitions and main results

A *graphon* is a symmetric measurable function $W: [0, 1]^2 \rightarrow [0, 1]$. We write

$$U := 1 - W.$$

For a bounded symmetric kernel $K: [0, 1]^2 \rightarrow \mathbb{R}$ and an integer $r \geq 2$, its cycle density is

$$t(C_r, K) := \int_{[0, 1]^r} \prod_{j=1}^r K(x_j, x_{j+1}) dx_1 \cdots dx_r, \quad x_{r+1} := x_1.$$

The same definition applies to the signed kernel $2W - 1$. For background on graphons, homomorphism densities, and the cut norm, see Lovász [3].

Definition 1.1 (Alternating-cycle density). *For $m \geq 2$, define*

$$\text{Alt}_{2m}(W) := \int_{[0, 1]^{2m}} \prod_{j=1}^m W(x_{2j-1}, x_{2j}) U(x_{2j}, x_{2j+1}) dx_1 \cdots dx_{2m},$$
$$x_{2m+1} := x_1.$$

Thus the colours alternate red, blue, red, blue around the cycle.

Theorem 1.2 (Strengthened alternating-cycle inequality). *Let $m \geq 3$ be odd. Then every graphon W satisfies*

$$4^m \text{Alt}_{2m}(W) + t(C_{2m}, 2W - 1) \leq 1. \quad (1)$$

Consequently,

$$\text{Alt}_{2m}(W) \leq 4^{-m}. \quad (2)$$

Equality in (2) holds if and only if $W = 1/2$ almost everywhere.

Writing $m = 2k + 1$ gives the usual formulation.

Corollary 1.3 (Alternating cycles of length $2 \pmod{4}$). *For every integer $k \geq 1$ and every graphon W ,*

$$\text{Alt}_{4k+2}(W) \leq 2^{-(4k+2)}.$$

Equality holds if and only if $W = 1/2$ almost everywhere.

Alternating cycles belong to the semi-inducibility framework of Basit, Granet, Horsley, Kündgen, and Staden [1], who proved sharp or nearly sharp results for alternating cycles of length divisible by four; the lengths congruent to 2 modulo 4 were left open. The first such case, length six, was settled by Chen and Noel [2] using flag algebras. Corollary 1.3 recovers that case ($k = 1$) and answers the question for the entire congruence class.

The strengthened inequality also gives a quantitative bound on quasirandomness. We use the cut norm

$$\|K\|_{\square} := \sup_{S, T \subseteq [0,1] \text{ measurable}} \left| \int_{S \times T} K(x, y) dx dy \right|.$$

Corollary 1.4 (Stability). *Let $m \geq 3$ be odd and put*

$$\delta := 1 - 4^m \text{Alt}_{2m}(W).$$

Then

$$\left\| W - \frac{1}{2} \right\|_{\square} \leq \frac{1}{2} \delta^{1/(2m)}. \quad (3)$$

We first reduce the graphon theorem to a finite-dimensional matrix inequality, then prove that inequality by a Schur-complement expansion.

2 Proof of the main results

We include the approximation argument in full so that every determinant appearing later is an ordinary finite-dimensional determinant.

Throughout this section we use the following dyadic approximation scheme. For $n \geq 1$, divide $[0, 1]$ into the dyadic intervals

$$I_{n,a} := \left[\frac{a}{2^n}, \frac{a+1}{2^n} \right), \quad 0 \leq a < 2^n,$$

with the convention that the last interval is closed, and let $\mathcal{A}_n$ be the finite sigma-algebra generated by these intervals. For a bounded measurable kernel K on $[0, 1]^2$, set

$$K_n := \mathbb{E}[K \mid \mathcal{A}_n \otimes \mathcal{A}_n].$$

Then K_n is a step kernel, constant on the rectangles $I_{n,a} \times I_{n,b}$. Conditional expectation preserves pointwise bounds, so $\|K_n\|_\infty \leq \|K\|_\infty$, and $0 \leq K \leq 1$ implies $0 \leq K_n \leq 1$. If K is symmetric, then so is K_n , because $\mathcal{A}_n \otimes \mathcal{A}_n$ is invariant under interchanging the two coordinates. Finally, the spaces of dyadic rectangle step functions are nested and their union is dense in $L^2([0, 1]^2)$; since K_n is the orthogonal projection of K onto the n th such space,

$$\|K_n - K\|_2 \longrightarrow 0, \quad \|K_n - K\|_1 \leq \|K_n - K\|_2 \longrightarrow 0, \quad (4)$$

where the last inequality uses that $[0, 1]^2$ has measure one.

Lemma 2.1 (Step-graphon reduction). *Fix $m \geq 2$. To prove either (1) or (2) for all W , it is enough to prove it for symmetric step graphons.*

Proof. Apply the dyadic scheme to W itself: the kernels

$$W_n := \mathbb{E}[W \mid \mathcal{A}_n \otimes \mathcal{A}_n]$$

are symmetric step graphons with $0 \leq W_n \leq 1$, and (4) gives $\|W_n - W\|_1 \rightarrow 0$.

We first check the alternating density. For each point $(x_1, \dots, x_{2m})$, the integrand of $\text{Alt}_{2m}(W_n)$ is a product of $2m$ factors, each equal either to $W_n(x_i, x_{i+1})$ or to $1 - W_n(x_i, x_{i+1})$. The corresponding factor for W differs by $\pm(W_n - W)(x_i, x_{i+1})$. The telescoping identity for two products, together with the fact that every factor lies in $[0, 1]$, gives

$$|\text{Alt}_{2m}(W_n) - \text{Alt}_{2m}(W)| \leq 2m\|W_n - W\|_1 \longrightarrow 0.$$

Next put

$$K_n := 2W_n - 1, \quad K := 2W - 1.$$

All factors in the cycle integrands have absolute value at most 1, so the same telescoping argument gives

$$|t(C_{2m}, K_n) - t(C_{2m}, K)| \leq 2m\|K_n - K\|_1 = 4m\|W_n - W\|_1 \longrightarrow 0.$$

Thus every inequality proved for W_n passes to W . □

We next describe the finite-dimensional operator model for step kernels. Let

$$[0, 1] = B_1 \sqcup \dots \sqcup B_N$$

be a finite measurable partition into sets of positive measure, and put

$$E := \text{span}\{\mathbf{1}_{B_1}, \dots, \mathbf{1}_{B_N}\} \subseteq L^2([0, 1]).$$

Every integral operator associated with a bounded kernel constant on the rectangles $B_a \times B_b$ maps $L^2([0, 1])$ into E and vanishes on $E^\perp$. Consequently, all its nonzero action occurs on the finite-dimensional real Hilbert space E .

For such a step kernel K , let T_K be the integral operator

$$(T_K f)(x) := \int_0^1 K(x, y) f(y) dy.$$

In the orthonormal basis

$$e_a := \frac{\mathbf{1}_{B_a}}{\sqrt{|B_a|}},$$

the matrix of $T_K|_E$ has entries

$$\langle e_a, T_K e_b \rangle = K_{ab} \sqrt{|B_a| |B_b|}.$$

It follows by direct multiplication that, for any sequence $K_1, \dots, K_r$ of step kernels constant on these rectangles,

$$\mathrm{Tr}(T_{K_1} \cdots T_{K_r}) = \int_{[0,1]^r} \prod_{j=1}^r K_j(x_j, x_{j+1}) dx_1 \cdots dx_r, \quad x_{r+1} := x_1. \quad (5)$$

In particular,

$$\mathrm{Tr}(T_K^r) = t(C_r, K). \quad (6)$$

Lemma 2.2 (Even signed cycles are nonnegative). *Let $K: [0, 1]^2 \rightarrow \mathbb{R}$ be a bounded symmetric measurable kernel, and let T_K be its integral operator on $L^2([0, 1])$. For every integer $r \geq 1$,*

$$t(C_{2r}, K) = \mathrm{Tr}(T_K^{2r}) = \|T_K^r\|_{\mathrm{HS}}^2 \geq 0. \quad (7)$$

Proof. First suppose that K is a symmetric step kernel. Then T_K has finite rank, the trace formula (6) gives $t(C_{2r}, K) = \mathrm{Tr}(T_K^{2r})$, and self-adjointness gives

$$\mathrm{Tr}(T_K^{2r}) = \mathrm{Tr}((T_K^r)^* T_K^r) = \|T_K^r\|_{\mathrm{HS}}^2.$$

For a general bounded symmetric kernel K , let $K_n = \mathbb{E}[K \mid \mathcal{A}_n \otimes \mathcal{A}_n]$ be its dyadic step approximations from the beginning of this section. They are symmetric, and (4) gives

$$\|K_n - K\|_2 \rightarrow 0, \quad \|K_n - K\|_1 \rightarrow 0.$$

Equivalently,

$$\|T_{K_n} - T_K\|_{\mathrm{HS}} \rightarrow 0.$$

The operators are uniformly bounded in operator norm, since $\|T_{K_n}\|_{\mathrm{op}} \leq \|T_{K_n}\|_{\mathrm{HS}} \leq \|K\|_2$ and similarly for T_K . The telescoping identity

$$T_{K_n}^r - T_K^r = \sum_{j=0}^{r-1} T_{K_n}^j (T_{K_n} - T_K) T_K^{r-1-j}$$

and the ideal inequality $\|ABC\|_{\mathrm{HS}} \leq \|A\|_{\mathrm{op}} \|B\|_{\mathrm{HS}} \|C\|_{\mathrm{op}}$ show that $T_{K_n}^r \rightarrow T_K^r$ in Hilbert–Schmidt norm. Hence

$$\|T_{K_n}^r\|_{\mathrm{HS}}^2 \longrightarrow \|T_K^r\|_{\mathrm{HS}}^2.$$

On the other hand, the same product-telescoping estimate as in Lemma 2.1, now with $M := \|K\|_\infty$ and $\|K_n\|_\infty \leq M$, gives

$$|t(C_{2r}, K_n) - t(C_{2r}, K)| \leq 2r M^{2r-1} \|K_n - K\|_1 \longrightarrow 0.$$

Passing to the limit in the step-kernel identity therefore gives

$$t(C_{2r}, K) = \|T_K^r\|_{\mathrm{HS}}^2.$$

Finally, T_K is Hilbert–Schmidt and hence bounded, so the ideal inequality gives

$$\|T_K^r\|_{\mathrm{HS}} \leq \|T_K\|_{\mathrm{op}}^{r-1} \|T_K\|_{\mathrm{HS}} < \infty.$$

Therefore T_K^r is Hilbert–Schmidt. Since T_K is self-adjoint,

$$T_K^{2r} = (T_K^r)^* T_K^r$$

is a positive trace-class operator, and

$$\mathrm{Tr}(T_K^{2r}) = \|T_K^r\|_{\mathrm{HS}}^2.$$

Combining the last two identities proves (7). $\square$

Now fix a symmetric step graphon W and choose a partition as above such that W is constant on every rectangle $B_a \times B_b$. The constant function $e := \mathbf{1}$ belongs to E and has norm one. Let

$$P := e \otimes e,$$

so that

$$Pf = \langle e, f \rangle e.$$

This is the rank-one orthogonal projection onto the constants, and it is also the operator T_1 associated with the constant kernel 1. Finally set

$$X := T_{2W-1}.$$

Since $2W - 1$ is symmetric, X is self-adjoint. Moreover,

$$T_W = \frac{P + X}{2}, \quad T_U = \frac{P - X}{2}. \quad (8)$$

By (5),

$$\mathrm{Alt}_{2m}(W) = \mathrm{Tr}((T_W T_U)^m) = 4^{-m} \mathrm{Tr}(((P + X)(P - X))^m). \quad (9)$$

Also,

$$\mathrm{Tr}(X^2) = \int_{[0,1]^2} (2W(x, y) - 1)^2 dx dy \leq 1. \quad (10)$$

Thus Theorem 1.2 follows from the following abstract matrix theorem.

Theorem 2.3 (Abstract alternating trace inequality). *Let H be a finite-dimensional real Hilbert space, let $e \in H$ be a unit vector, and let*

$$P = e \otimes e.$$

Let $X = X^$ satisfy*

$$\tau := \mathrm{Tr}(X^2) \leq 1.$$

For every odd positive integer m ,

$$\mathrm{Tr}(((P + X)(P - X))^m) + \mathrm{Tr}(X^{2m}) \leq 1. \quad (11)$$

The proof is given in the remainder of this section. Put

$$L := (P + X)(P - X). \quad (12)$$

Although L need not be self-adjoint, the determinant–trace identity used below is valid for every finite matrix.

All power-series identities below may be read either formally or as Taylor expansions for z sufficiently close to zero. Since all matrices are finite-dimensional, the two interpretations agree coefficient by coefficient.

Define

$$D(z) := I + zX^2, \quad N(z) := D(z)^{-1}, \quad u := Xe.$$

Also define the scalar functions

$$h(z) := \langle e, N(z)e \rangle, \tag{13}$$

$$k(z) := \langle e, XN(z)e \rangle = \langle u, N(z)e \rangle, \tag{14}$$

$$\ell(z) := \langle e, X^2N(z)e \rangle = \langle u, N(z)u \rangle. \tag{15}$$

Since

$$N(z) + zX^2N(z) = I,$$

we have

$$h(z) + z\ell(z) = 1. \tag{16}$$

Lemma 2.4 (Rank-two determinant factorisation). *With L as in (12),*

$$\det(I - zL) = \det(I + zX^2)(1 - zF(z)), \tag{17}$$

where

$$F(z) := h(z)^2 + zk(z)^2. \tag{18}$$

Proof. Expanding L gives

$$L = P - PX + XP - X^2.$$

Since $PX = e \otimes u$ and $XP = u \otimes e$,

$$\begin{aligned} I - zL &= I + zX^2 - zP + zPX - zXP \\ &= D(z) + ze \otimes (u - e) - zu \otimes e. \end{aligned}$$

Introduce the maps $\mathcal{U}: \mathbb{R}^2 \rightarrow H$ and $\mathcal{V}: \mathbb{R}^2 \rightarrow H$ whose columns are

$$\mathcal{U} = \begin{bmatrix} ze & -zu \end{bmatrix}, \quad \mathcal{V} = \begin{bmatrix} u - e & e \end{bmatrix}.$$

Then

$$I - zL = D(z) + \mathcal{U}\mathcal{V}^*.$$

The matrix determinant lemma gives

$$\det(I - zL) = \det D(z) \det(I_2 + \mathcal{V}^*N(z)\mathcal{U}).$$

Because $N(z)$ is a function of X^2 , it commutes with X . Using (13)–(15), we obtain

$$I_2 + \mathcal{V}^*N(z)\mathcal{U} = \begin{pmatrix} 1 + z(k - h) & z(k - \ell) \\ zh & 1 - zk \end{pmatrix}.$$

Its determinant is

$$\begin{aligned} &(1 + z(k - h))(1 - zk) - z^2h(k - \ell) \\ &= 1 - zh - z^2k^2 + z^2h\ell. \end{aligned}$$

Using (16), namely $z\ell = 1 - h$, this becomes

$$1 - zh^2 - z^2k^2 = 1 - z(h^2 + zk^2).$$

Substitution proves (17). □

Diagonalise X in an orthonormal eigenbasis:

$$Xv_i = \lambda_i v_i, \quad \omega_i := |\langle e, v_i \rangle|^2.$$

Eigenvalues are listed with multiplicity. Then

$$\omega_i \geq 0, \quad \sum_i \omega_i = 1, \quad \sum_i \lambda_i^2 = \tau \leq 1. \quad (19)$$

The scalar resolvents are

$$h(z) = \sum_i \frac{\omega_i}{1 + z\lambda_i^2}, \quad k(z) = \sum_i \frac{\omega_i \lambda_i}{1 + z\lambda_i^2}. \quad (20)$$

Consequently,

$$F(z) = \sum_{i,j} \omega_i \omega_j \frac{1 + z\lambda_i \lambda_j}{(1 + z\lambda_i^2)(1 + z\lambda_j^2)}. \quad (21)$$

For $n \geq 0$, define the polynomial

$$c_n(x, y) := \frac{x^{2n+1} + y^{2n+1}}{x + y}, \quad (22)$$

where the quotient is interpreted by continuity at $x + y = 0$. Equivalently,

$$c_n(x, y) = \sum_{r=0}^{2n} (-1)^r x^{2n-r} y^r. \quad (23)$$

In particular,

$$c_n(x, -x) = (2n + 1)x^{2n}.$$

Lemma 2.5 (Positivity and recurrence for c_n). *For all real x, y and all $n \geq 0$,*

$$c_n(x, y) \geq 0. \quad (24)$$

Moreover, for $n \geq 1$,

$$(x^2 + y^2)c_n(x, y) - c_{n+1}(x, y) = x^2 y^2 c_{n-1}(x, y). \quad (25)$$

Proof. Let $f(t) = t^{2n+1}$. If $x + y \neq 0$, then

$$c_n(x, y) = \frac{f(x) - f(-y)}{x - (-y)}.$$

The function f is increasing on $\mathbb{R}$, so this difference quotient, the slope of a chord of f , is nonnegative. If $x = -y$, the continuous value is $f'(x) = (2n + 1)x^{2n} \geq 0$. This proves (24).

For the recurrence, multiply the claimed identity by $x + y$. The left side becomes

$$\begin{aligned} & (x^2 + y^2)(x^{2n+1} + y^{2n+1}) - (x^{2n+3} + y^{2n+3}) \\ &= x^2 y^{2n+1} + y^2 x^{2n+1} \\ &= x^2 y^2 (x^{2n-1} + y^{2n-1}), \end{aligned}$$

which is $(x + y)x^2 y^2 c_{n-1}(x, y)$. Since both sides are polynomials, the identity holds also when $x + y = 0$. $\square$

The partial-fraction identity

$$\frac{1 + zxy}{(1 + zx^2)(1 + zy^2)} = \frac{x}{x + y} \frac{1}{1 + zx^2} + \frac{y}{x + y} \frac{1}{1 + zy^2} \quad (26)$$

for $x + y \neq 0$, extended continuously to $x + y = 0$, gives

$$\frac{1 + zxy}{(1 + zx^2)(1 + zy^2)} = \sum_{n \geq 0} (-1)^n c_n(x, y) z^n. \quad (27)$$

Combining (21) and (27), write

$$F(z) = \sum_{n \geq 0} (-1)^n \beta_n z^n, \quad (28)$$

where

$$\beta_n := \sum_{i,j} \omega_i \omega_j c_n(\lambda_i, \lambda_j). \quad (29)$$

Lemma 2.6 (Monotonicity of the coefficients β_n). *The sequence $(\beta_n)_{n \geq 0}$ satisfies*

$$1 = \beta_0 \geq \beta_1 \geq \beta_2 \geq \cdots \geq 0. \quad (30)$$

More precisely,

$$\beta_{n+1} \leq \tau \beta_n \quad (n \geq 1). \quad (31)$$

Proof. Positivity follows from Lemma 2.5. Since $c_0 \equiv 1$ and $\sum_i \omega_i = 1$, we have $\beta_0 = 1$.

We first prove $\beta_1 \leq \tau$. Put

$$\mu_r := \sum_i \omega_i \lambda_i^r = \langle e, X^r e \rangle.$$

Since

$$c_1(x, y) = x^2 - xy + y^2,$$

we obtain

$$\beta_1 = 2\mu_2 - \mu_1^2. \quad (32)$$

Decompose $H = \mathbb{R}e \oplus e^\perp$ and write

$$X = \begin{pmatrix} a & w^* \\ w & A \end{pmatrix}.$$

Then

$$\mu_1 = a, \quad \mu_2 = a^2 + \|w\|^2,$$

and hence

$$\beta_1 = a^2 + 2\|w\|^2. \quad (33)$$

On the other hand,

$$\tau = \text{Tr}(X^2) = a^2 + 2\|w\|^2 + \text{Tr}(A^2). \quad (34)$$

Thus $\beta_1 \leq \tau \leq 1$.

Now let $n \geq 1$. For a diagonal pair $i = j$, one has

$$c_n(\lambda_i, \lambda_i) = \lambda_i^{2n},$$

so

$$c_{n+1}(\lambda_i, \lambda_i) = \lambda_i^2 c_n(\lambda_i, \lambda_i) \leq \tau c_n(\lambda_i, \lambda_i).$$

For $i \neq j$, Lemma 2.5 gives

$$\begin{aligned} c_{n+1}(\lambda_i, \lambda_j) &\leq (\lambda_i^2 + \lambda_j^2) c_n(\lambda_i, \lambda_j) \\ &\leq \tau c_n(\lambda_i, \lambda_j), \end{aligned}$$

where the last inequality follows because the eigenvalues are listed with multiplicity and

$$\lambda_i^2 + \lambda_j^2 \leq \sum_s \lambda_s^2 = \tau.$$

Averaging with the nonnegative weights $\omega_i \omega_j$ proves (31). Since $\tau \leq 1$, the sequence is nonincreasing. $\square$

The following purely one-dimensional lemma isolates the role of the parity of m .

Lemma 2.7 (Odd logarithmic coefficient). *Let*

$$1 = \beta_0 \geq \beta_1 \geq \beta_2 \geq \cdots$$

and put

$$F(z) := \sum_{n \geq 0} (-1)^n \beta_n z^n.$$

Then, for every odd positive integer m ,

$$m[z^m](-\log(1 - zF(z))) \leq 1. \quad (35)$$

More precisely, if

$$d_n := \beta_n - \beta_{n+1} \geq 0, \quad G(z) := \sum_{n \geq 0} d_n z^n,$$

then

$$1 - m[z^m](-\log(1 - zF(z))) = m \sum_{r=1}^{(m-1)/2} \frac{1}{r} [z^{m-2r}] G(z)^r. \quad (36)$$

In particular, the right side is nonnegative.

Proof. The coefficient of z^n in $(1+z)F(z)$ is the successive difference of the β_j with alternating sign. Explicitly,

$$(1+z)F(z) = 1 + zG(-z). \quad (37)$$

Therefore

$$\begin{aligned} (1+z)(1 - zF(z)) &= 1 + z - z(1+z)F(z) \\ &= 1 - z^2 G(-z). \end{aligned} \quad (38)$$

Taking formal logarithms gives

$$-\log(1 - zF(z)) = \log(1+z) - \log(1 - z^2 G(-z)). \quad (39)$$

For odd m ,

$$[z^m] \log(1+z) = \frac{1}{m}.$$

Furthermore,

$$-\log(1 - z^2 G(-z)) = \sum_{r \geq 1} \frac{z^{2r} G(-z)^r}{r}. \quad (40)$$

If m is odd, then $m - 2r$ is odd. Since G has nonnegative coefficients,

$$[z^{m-2r}]G(-z)^r = -[z^{m-2r}]G(z)^r \leq 0.$$

Substituting this into (39) gives the exact identity (36), and hence (35). $\square$

We use the standard finite-dimensional identity

$$-\log \det(I - zM) = \sum_{r \geq 1} \frac{\text{Tr}(M^r)}{r} z^r \quad (41)$$

for an arbitrary finite matrix M . For completeness, differentiating the left side and using Jacobi's formula gives

$$\frac{d}{dz} [-\log \det(I - zM)] = \text{Tr}(M(I - zM)^{-1}) = \sum_{r \geq 0} \text{Tr}(M^{r+1}) z^r.$$

Both sides of (41) vanish at $z = 0$, proving the identity coefficientwise. No diagonalizability of M is required.

Proof of Theorem 2.3. Take minus the logarithm of the determinant factorisation (17):

$$-\log \det(I - zL) = -\log \det(I + zX^2) - \log(1 - zF(z)). \quad (42)$$

By (41),

$$-\log \det(I - zL) = \sum_{r \geq 1} \frac{\text{Tr}(L^r)}{r} z^r.$$

Since X is self-adjoint,

$$-\log \det(I + zX^2) = \sum_{r \geq 1} \frac{(-1)^r \text{Tr}(X^{2r})}{r} z^r. \quad (43)$$

Let m be odd. Comparing the coefficient of z^m in (42) gives

$$\text{Tr}(L^m) = -\text{Tr}(X^{2m}) + m[z^m](-\log(1 - zF(z))). \quad (44)$$

By Lemma 2.6, the coefficients of F satisfy the hypotheses of Lemma 2.7. Hence

$$m[z^m](-\log(1 - zF(z))) \leq 1.$$

Substitution into (44) proves

$$\text{Tr}(L^m) + \text{Tr}(X^{2m}) \leq 1.$$

This is (11). $\square$

The proof actually supplies an exact formula for the gap.

Proposition 2.8 (Exact formula for the gap). *With the notation above, let*

$$d_n := \beta_n - \beta_{n+1}, \quad G(z) := \sum_{n \geq 0} d_n z^n.$$

For every odd positive integer m ,

$$1 - \text{Tr}(L^m) - \text{Tr}(X^{2m}) = m \sum_{r=1}^{(m-1)/2} \frac{1}{r} [z^{m-2r}] G(z)^r. \quad (45)$$

Every term on the right is nonnegative.

Proof. Combine (44) with the exact identity (36). □

Proof of Theorem 1.2. By Lemma 2.1, first assume that W is a symmetric step graphon. Work on the finite-dimensional space E from the finite-rank reduction, and define P and X as in (8). The bound (10) gives

$$\text{Tr}(X^2) \leq 1.$$

Therefore Theorem 2.3, together with (9) and (6), yields

$$4^m \text{Alt}_{2m}(W) + t(C_{2m}, 2W - 1) = \text{Tr}(((P + X)(P - X))^m) + \text{Tr}(X^{2m}) \leq 1.$$

This proves (1) for step graphons. The approximation in Lemma 2.1 passes it to every graphon.

Lemma 2.2, applied to the signed kernel $2W - 1$, gives

$$t(C_{2m}, 2W - 1) = \text{Tr}(X^{2m}) = \|X^m\|_{\text{HS}}^2 \geq 0.$$

Dropping this nonnegative term from (1) proves (2).

Suppose equality holds in (2). Then (1) and Lemma 2.2 give

$$0 \leq \text{Tr}(X^{2m}) = \|X^m\|_{\text{HS}}^2 \leq 1 - 4^m \text{Alt}_{2m}(W) = 0.$$

Hence $X^m = 0$. Because X is self-adjoint,

$$\|X\|_{\text{op}}^m = \|X^m\|_{\text{op}} = 0,$$

so $X = 0$. The Hilbert–Schmidt kernel of X is $2W - 1$, and therefore $2W - 1 = 0$ almost everywhere. Hence $W = 1/2$ almost everywhere. Conversely, $W \equiv 1/2$ gives

$$\text{Alt}_{2m}(W) = 2^{-2m} = 4^{-m},$$

so equality does occur. □

Proof of Corollary 1.4. Put $X = T_{2W-1}$. The strengthened inequality gives

$$\text{Tr}(X^{2m}) \leq \delta.$$

By (2), $\delta \geq 0$. If (λ_j) are the eigenvalues of the compact self-adjoint operator X , counted with multiplicity, then

$$\|X\|_{\text{op}}^{2m} = \sup_j |\lambda_j|^{2m} \leq \sum_j |\lambda_j|^{2m} = \text{Tr}(X^{2m}) \leq \delta.$$

For measurable $S, T \subseteq [0, 1]$,

$$\left| \int_{S \times T} (2W - 1) \right| = |\langle \mathbf{1}_S, X \mathbf{1}_T \rangle| \leq \|X\|_{\text{op}} \|\mathbf{1}_S\|_2 \|\mathbf{1}_T\|_2 \leq \|X\|_{\text{op}}.$$

Hence

$$\|2W - 1\|_{\square} \leq \|X\|_{\text{op}} \leq \delta^{1/(2m)}.$$

Dividing by 2 proves (3). $\square$

3 Remarks

Although the theorem is an unrestricted semi-inducibility result, the signed-cycle term also gives a simple density-sensitive bound.

Corollary 3.1 (A fixed-edge-density upper bound). *Let*

$$p := \int_{[0,1]^2} W(x, y) dx dy.$$

For every odd $m \geq 3$,

$$\text{Alt}_{2m}(W) \leq 4^{-m} (1 - (2p - 1)^{2m}). \quad (46)$$

Proof. For $X = T_{2W-1}$ and $e = \mathbf{1}$,

$$\langle e, X e \rangle = 2p - 1.$$

Let ν be the spectral probability measure of X at e . Since $t \mapsto t^{2m}$ is convex,

$$\langle e, X^{2m} e \rangle = \int t^{2m} d\nu(t) \geq \left(\int t d\nu(t) \right)^{2m} = (2p - 1)^{2m}.$$

Also X^{2m} is positive semidefinite, so

$$\text{Tr}(X^{2m}) \geq \langle e, X^{2m} e \rangle.$$

Substitute this into (1). $\square$

The inequalities (1) and (2), together with the fact that $W \equiv 1/2$ attains the equality, have been formalised in the Lean 4 proof assistant [5], on top of the mathlib library [4]; the development accompanies this note. Corollaries 1.4 and 3.1 and the uniqueness of the equality case are not formalised. The formal proof differs from the one given here in three ways.

First, instead of the step-graphon approximation (Lemma 2.1), we write the graphon quantity $4^m \text{Alt}_{2m}(W) + t(C_{2m}, 2W - 1)$ and its matrix counterpart as the same polynomial in the moments $\mu_g = \langle \mathbf{1}, X^g \mathbf{1} \rangle$, $g < 2m$, and compress X to the Krylov subspace $\text{span}\{\mathbf{1}, X\mathbf{1}, \dots, X^{2m}\mathbf{1}\}$; this produces a finite symmetric matrix with the same moments and $\text{Tr}(A^2) \leq 1$, to which Theorem 2.3 applies directly.

Second, the formal proof never forms a logarithm or a determinant. Instead of handling $m[z^m](-\log A(z))$, it uses the operator $\Lambda(A) := -zA'(z)/A(z)$: since $z d/dz$ multiplies the coefficient of z^m by m , one has $[z^m]\Lambda(A) = m[z^m](-\log A(z))$, while Λ itself involves no logarithm. The rule $\Lambda(fg) = \Lambda(f) + \Lambda(g)$, immediate from $(fg)' = f'g + fg'$, replaces the only property of the logarithm that the argument uses. Similarly, the logarithm in (44) is read as $[z^m]\Lambda(1 - zF(z))$, and the identity is proved not through (41) but directly from the rank-two structure in the proof of Lemma 2.4, as a computation with the 2×2 matrix $I_2 + \mathcal{V}^* N(z) \mathcal{U}$; the only determinants that appear are 2×2 .

Third, in Lemma 2.6, we obtain $\beta_1 \leq \tau$ from a direct sum-of-squares identity, without decomposing H along e .

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